Introduction

Data analysis and vizualization in Python - RSE course

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RSE = Research software engineering

- **Software development & maintenance** for research projects
- **Training and workshops** on programming and research software skills
- **Consultations** for research projects and grant proposals
- **Support for HPC computing** (e.g. Chimera cluster)
- **Development of research tools and platforms** (e.g. AI Sandbox)
- contact: rse.cuni.cz
- MY ROLE:
    - focus on LLMs, AI in general and AI agents
    - finished master studies at CTU - Software engineering (Artificial intelligence)
- YOUR ROLE?

Course overview

- *1-day hands-on practical* Data analysis and vizualization in Python *for beginners & researchers*
- **Course time plan**
- **sticky notes**

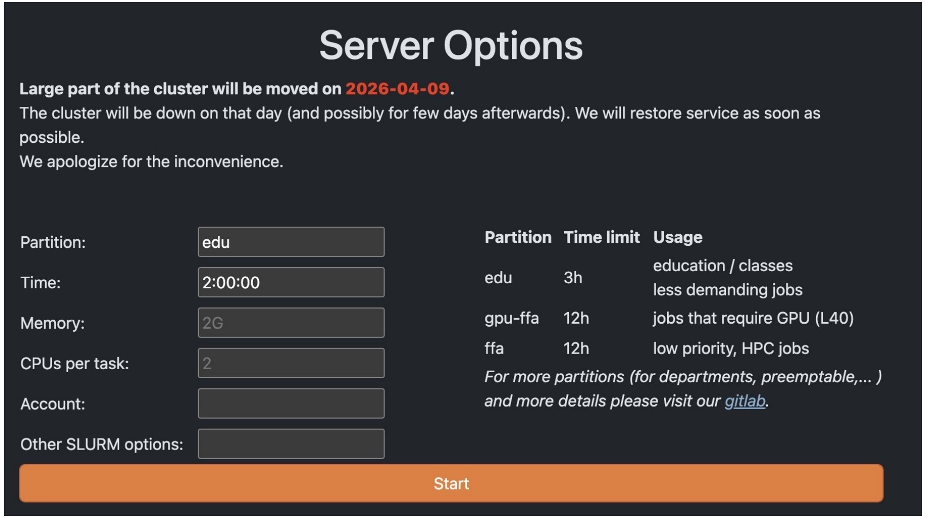
13:10 – 13:30	Introduction & setup
13:30 – 14:00	loading datasets & dataframes
14:00 – 14:40	dataframes filtering & aggregation
14:40 – 15:00	break
15:00 – 15:30	errors in data
15:30 – 16:30	Matplotlib
16:30 – 17:00	project

Course materials

- Jupyter notebooks and slides on github

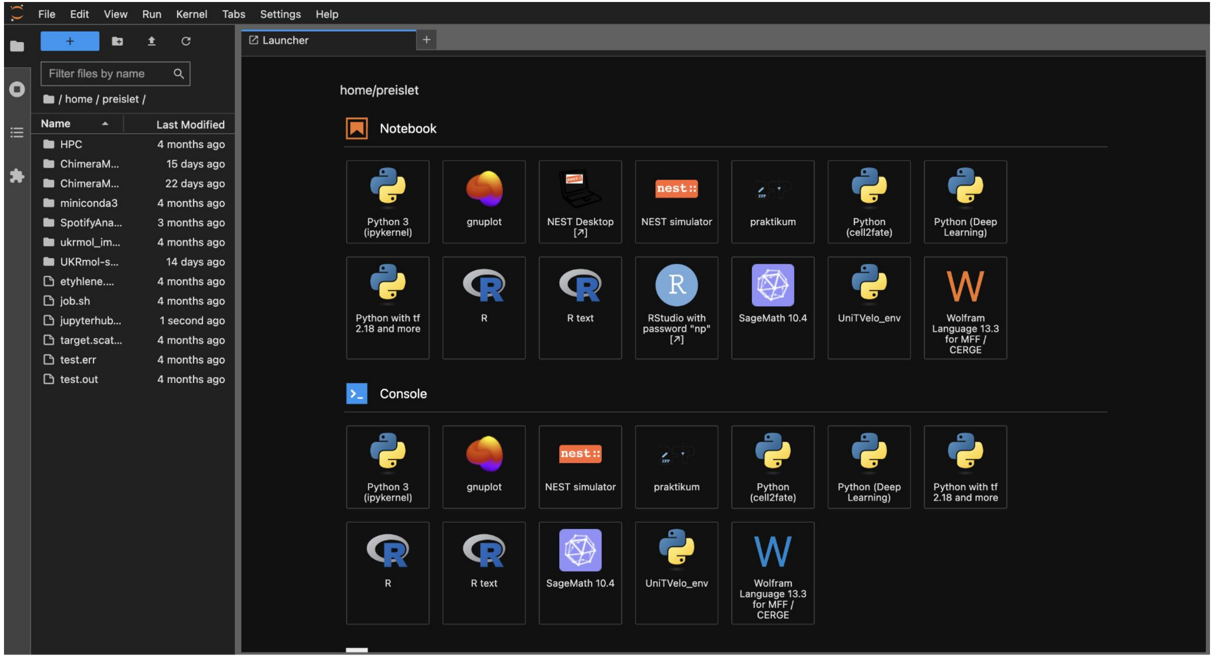
https://github.com/rse-cuni/training-materials/tree/main/Data-analysis-Python

- Jupyter notebooks available on chimera
- connect to chimera:
    - https://hpc.troja.mff.cuni.cz:8000
    - You will be asked for credentials.
    - Put in your SIS login name and password
    - You should end up here:
    - Press Start



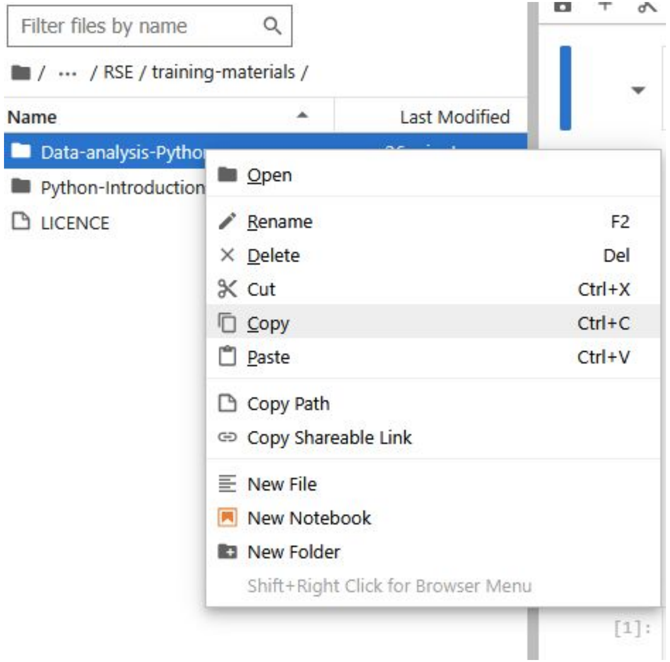
Chimera JupyterLab setup

- go to /home/teaching/RSE/training-materials



Chimera JupyterLab setup

- right-click on Data-analysis-Python and click Copy



Chimera JupyterLab setup

- go back to your home directory:

  /home/your_cas_login

- and right click and Paste